CroMoDa: Unsupervised Oriented SAR Ship Detection via Cross-Modality Distribution Alignment

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***Abstract*—Most state-of-the-art synthetic aperture radar (SAR) ship detection methods based on deep learning require large amounts of labeled data for network training. However, the annotation process requires significant manpower and resources especially for SAR images, since relevant background knowledge should be necessary for the annotators. Considering the available optical imagery datasets with labels, we propose an unsupervised oriented SAR ship detection method based on cross-modality distribution alignment, termed as CroMoDa. It consists of four components: 1) image-level feature alignment; 2) low-level feature despeckling; 3) cross-modality pseudo-label self-training; and 4) cross-modality object alignment. By aligning the multilevel feature distributions, modality-invariant features across the two imagery modalities can be learned. Considering speckle noise and other interferences in SAR images, the proposed loss term conducted on low-level features can enhance the object information to improve the detection accuracy. Moreover, the proposed pseudo-label self-training can better generate oriented SAR ship annotations than the other methods, which facilitates learning more discriminative features for SAR ship instances. With two optical–SAR dataset configurations, the proposed method is effectively evaluated by comparing to the other state of the arts, which demonstrates the great potential for SAR ship detection in real applications.**

***Index Terms*—Cross modality, synthetic aperture radar (SAR), synthetic aperture radar ship detection, transfer learning, unsupervised object detection.**

I. Introduction

SYNTHETIC aperture radar (SAR) is widely used in earth observation, military reconnaissance, and ocean monitoring due to its ability to provide high-resolution remote sensing images under all-weather and all-day conditions. SAR images, due to their unique imaging mechanism, have the capability to acquire images under adverse weather conditions compared to optical imaging. This makes SAR images show great potential in applications such as ship detection and tracking [1], [2], [3], [4], [5].

Traditional SAR ship detection methods mainly include constant false alarm rate (CFAR) [6], [7], [8] and extended fractal (EF) [9]. CFAR is essentially based on a segmentation idea, where the detection threshold is compared with the grayscale value of pixels, treating pixels above the threshold as ships and those below it as backgrounds. The detection threshold is determined by the statistical characteristics of the background clutter, and calculating this threshold often requires significant computational resources, resulting in poor real-time performance and low detection accuracy in complex background. EF fully integrates the grayscale information of the image and the spatial distribution of grayscale values, using the spatial difference in reflection energy between the ship and clutter for ship detection. However, EF requires high image quality, and the presence of noise, blur, or distortion in the image can affect the extraction and analysis of fractal features, thereby impacting the accuracy and stability of ship detection. In recent years, with the rapid development of deep learning methods, a large number of approaches based on convolutional neural networks (CNNs) have achieved significant results in the field of SAR ship detection. Currently, detectors based on CNNs are mainly divided into single-stage detection algorithms [10], [11], [12], [13] and two-stage detection algorithms [14], [15], [16], [17]. Single-stage detection algorithms have advantages in computational speed compared with two-stage detection ones, but are inferior in terms of detection accuracy [18]. Faster region-based convolutional neural network (Faster R-CNN) is a typical two-stage detector that generates candidate boxes through a region proposal network (RPN) and then processes these boxes with classification and regression branches, achieving superior performance [16].

Although SAR ship detection methods based on CNNs have achieved significant improvements in performance, they inevitably require a large amount of labeled data for network training. Currently, obtaining sufficient annotated data for supervised training still faces several challenges. On the one hand, there are few instance-level annotated datasets available for SAR ship detection tasks, and even fewer datasets adopt oriented annotations. Moreover, most of the labeled datasets are small,

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and training networks on them can lead to an overfitting problem, reducing generalization performance. On the other hand, the interpretation of SAR images requires annotators with relevant professional knowledge, further increasing the manpower and time costs. In contrast, the annotation of optical images is relatively easier, making the use of optical images to train networks for SAR ship detection a new perspective. However, in traditional supervised deep learning, a network can only achieve good performance if both the training and testing data follow the same probability distribution. The distribution difference between optical and SAR images implies that networks trained on optical images cannot achieve satisfactory results on SAR images for inference [19], [20], [21]. This performance degradation caused by such difference is referred to as the domain shift problem [22]. The goal of this article is to achieve SAR ship detection by exploiting the labeled optical images and unlabeled SAR images themselves, where the unsupervised domain shift problem should be tackled.

Chen et al. [22] are the first to achieve cross-domain detection by aligning image-level and instance-level features. Recently, many subsequent works have been proposed on the basis of such a framework [23], [24], [25], [26], [27]. However, the aforementioned studies on domain-adaptive object detection are designed for natural images, which deal with domain-shift problems such as weather change, day and night change, etc. Since the imaging mechanisms of optical and SAR images are totally different, such an image modality change will make the direct applications of those methods struggle with the detection performances of SAR ships. Building on adversarial domain adaptive learning, Zhao et al. [28] consider the significant differences in scattering intensities of SAR images and use entropy vectors to distinguish between high-entropy and low-entropy areas, to which corresponding weights are assigned. This method is used for cross-domain detection tasks with only SAR imagery and relies on a large amount of annotated SAR images. Xu et al. [29] attempt to gradually reduce the distribution differences between optical and SAR images through a multilevel alignment network at the image, convolution, and instance levels. Such feature alignment in cases of large distribution differences is prone to negative transfer, which limits the performance improvement. Shi et al. [30] propose an unsupervised domain adaptation detection method for the transition from optical to SAR images, transferring knowledge progressively at the pixel level, feature level, and inference level. At the pixel level, generative adversarial networks (GANs) are used to convert optical images into SAR-like images in the transition domain. At the feature level, domain-invariant features are acquired through adversarial learning. At the inference level, self-training is conducted using pseudo-labels generated by the feature-aligned detector. Although this method can achieve good detection results, the three independent stages can increase the training cost of the network, and the non-end-to-end framework might introduce cumulative errors. Pan et al. [31] note that two distributions with low correlation are more likely to encounter negative transfer, and introduce the prediction consistency loss [22] to solve such an issue.

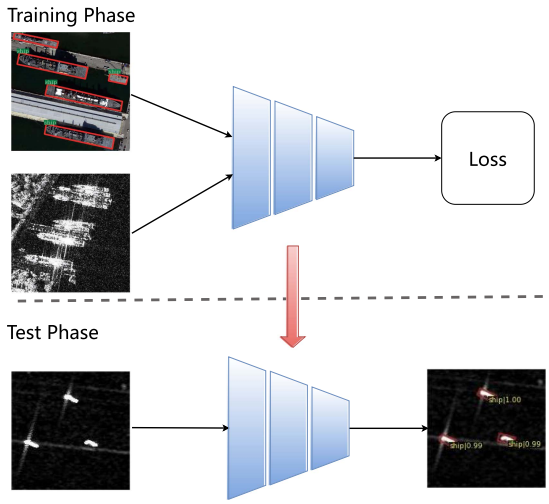


Fig. 1. Unsupervised oriented SAR ship detection network is learned by using labeled optical and unlabeled SAR images.

Although the abovementioned methods can mitigate the detection degradation problem caused by the significant modality change from optical to SAR imagery, they ignore the impact of speckle noise and other interference in SAR images on the detection performance. Moreover, without the supervised information from SAR images, situations where object features are aligned with the ones from the background may occur in existing methods, which reduces the discrimination capability of networks between objects and background areas. In addition, the oriented annotation of SAR ships is more suitable for applications in real scenarios due to the fewer effects produced by the sea clutter background than the horizontal annotation. Taking these into account, as shown in Fig. 1, we aim to achieve unsupervised oriented SAR ship detection given labeled optical images and SAR images without any supervised information, and propose a cross-modality distribution alignment method, named `CroMoDa`. Based on the Faster R-CNN structure, we propose four components to mitigate the impact caused by the modality change on the detection performance and achieve cross-modality-oriented SAR ship detection in an unsupervised manner. Specifically, they are composed of: 1) image-level feature alignment; 2) low-level feature despeckling; 3) cross-modality pseudo-label self-training; and 4) cross-modality object alignment. The image-level feature alignment is exploited for learning modality-invariant features at multiscales. By introducing low-level feature despeckling loss, the impact of speckle noise on low-level features can be prevented, as well as other interference produced by sea clutter and strong sidelobes. To learn more SAR-specific features, the cross-modality pseudo-label self-training strategy generates high-quality pseudo-labels to provide supervised information for SAR ships, which can reduce the occurrence of false alarms. The proposed cross-modality object alignment aims to precisely narrow the distribution distances between ship instances extracted from both modalities. To this end, the contributions of this article can be summarized as follows.

1) We propose an end-to-end unsupervised oriented SAR ship detection framework based on cross-modality distribution alignment. To the best of our knowledge, this is the first end-to-end unsupervised oriented SAR ship detection framework by knowledge transfer from optical to SAR images in this field.
2) Compared to other state of the arts (SOTAs), we consider the impact of speckle noise and other interference inherent in SAR images on cross-modality detection. The proposed low-level feature despeckling loss can mitigate noise and interference at the feature level while retaining the characteristics of SAR ships.
3) To enable the network to better capture ship features from SAR images, we introduce a cross-modality pseudo-label self-training strategy. Through the proposed forward twice backward once (FTBO) mechanism, we achieve end-to-end pseudo-label self-training. Furthermore, by introducing a pseudo-label bank, we enhance the network's training speed and avoid the impact of cumulative errors, which are commonly encountered in the pseudo-label learning stage.
4) The effectiveness of CroMoDa is evaluated on two dataset configurations by comparing it to the other SOTAs. On average, the detection improvement can be reached around 10% mean average precision (mAP) compared to the second-best method.

Notably, since most current cross-modality unsupervised object detection methods are based on Faster R-CNN, our method is also designed based on Faster R-CNN for ease of comparison. However, the method proposed in this article can be easily applied to other object detectors such as SSD [10], YOLO [11], RetinaNet [12], etc.

The rest of this article is organized as follows. Section II presents some related work from the perspective of SAR ship detection and transfer learning. Section III introduces the proposed CroMoDa in detail. Extensive experiments are conducted in Section IV with the analysis. Finally, Section V concludes this article and provides some hints on future research.

II. Related Work

A. SAR Ship Detection

Deep learning, with its powerful feature extraction and pattern recognition capabilities, has propelled SAR ship detection into a new era. Yang et al. [32] design a receptive field enhancement module that captures multiscale contextual information, improving the detection performance for ships of different sizes. Niu et al. [33] consider the high computational cost of feature pyramid networks in current detectors, innovatively design an encoder–decoder detection model that can extract multilevel features. They also exploit masks to estimate the orientation information of ships, using this information for weak supervision of the network. Considering the speckle noise in SAR images, Zhao et al. [34] employ a dual feature fusion attention mechanism to suppress noise background and integrate shallow features with denoised features, enhancing the network's robustness against speckle noise. Zhang et al. [35] propose a frequency attention module to adaptively process the frequency information of SAR images, which makes the network focus more on ships than sea clutter. To improve the detection speed, Wang et al. [36] encode the convolutional kernels of well-performed networks into a list of probabilities and use the firefly optimization algorithm to search for a set of lightweight network encodings. Through pruning techniques, they transform the network encoding into a lightweight ship detection network, achieving a balance between accuracy and speed. Zhang et al. [37] propose a multiscale global scattering feature association network for SAR ship detection. In SAR images, ships often have large aspect ratios and arbitrary orientations. The horizontal bounding box detection algorithms lead to redundant background areas, making them difficult to accurately locate targets in complex scenes. Chen et al. [38] propose an adaptive recalibration network for detecting multiscale arbitrarily oriented ships. Pan et al. [39] propose a multistage oriented detection network using an angle correlation strategy to generate multiangle anchor boxes. By progressively increasing the intersection over union (IoU) threshold for assigning positive and negative candidate boxes across three stages, they ensure accuracy in localization and classification. Yang et al. [40] introduce a dual feature alignment module for capturing ship orientation and shape information and design an adaptive IoU calculation method that incorporates the aspect ratio of ships. To enhance the efficiency of detection in large SAR images, Jia et al. [41] identify areas that may contain ships in very large images through sea–land segmentation and CFAR, then use detectors to identify ships within these areas, and finally develop a false alarm discrimination network to further eliminate false alarms, achieving rapid and high-accuracy SAR ship detection. Wu et al. [42] propose an instance-segmentation-assisted ship detection network, by thresholding segmentation results to obtain approximate ship contours and integrating the instance segmentation results into the detection pipeline to achieve high-accuracy ship detection. The aforementioned deep learning methods have achieved good results in SAR ship detection, while they all require the training and test datasets to have the same distribution. In real scenarios, it is challenging to require the test dataset to follow the same distribution, and the cost of reannotation is high.

B. Domain Adaptation for Object Detection

Domain adaptation has been widely applied in object detection. For similar images, changes in viewing angles, lighting conditions, and imaging methods can cause variations in feature distributions, making the development of a cross-domain detection model very important [43], [44], [45]. Currently, methods of unsupervised domain adaptation in the detection field generally employ three techniques: style transfer, feature alignment, and self-training. Among these, style transfer often serves as a preliminary task for feature alignment and self-training, transforming source-domain images into images with a style similar to the target domain to mitigate domain distribution differences. Style transfer is typically based on GANs [46], [47], [48]. Chen et al. [22] align the feature distributions between the source and target domains at the image and instance feature

levels using adversarial learning. This is the first time adversarial training used for cross-domain object detection tasks. Saito et al. [23] propose a strong–weak alignment model that tightly aligns local features such as texture and color, while applying weak alignment to global features to reduce the domain gap without compromising the model's performance. Chen et al. [26] point out that not all features have equal transferability and enhance global discriminability by reweighting interpolated image features. Wu et al. [27] design two disentangling layers to decompose features into domain-specific and domain-invariant features, and extract instance-invariant features for object detection. Cai et al. [49] propose a background-focused distribution alignment for pedestrian detection, which avoids the issue of misalignment between foreground features in source-domain images and background features in target domain images. Li et al. [50] integrate self-supervised learning tasks, such as rotation angle prediction and strong/weak data augmentation into cross-domain object detection, to facilitate knowledge transfer between domains. Zhou et al. [51] introduce a new class-level discriminator that achieves the discriminability of instances with different categories while also maintaining category consistency between source and target domains. For self-training methods, pseudo-labels in the target domain are generated and exploited for the supervised training of detection networks. Khodabandeh et al. [52] propose a robust object detection framework to address issues with incorrect category labels and inaccurate bounding box locations in pseudo-labels. Kim et al. [24] present a domain adaptive detection method based on single-stage object detectors, using weak self-training to reduce the impact of inaccurate pseudo-labels and leveraging adversarial background score regularization to extract discriminative features of backgrounds, thereby reducing domain shift. Li et al. [53] propose an adaptive teacher framework, where the student network uses domain adversarial training to generate domain-invariant features, and exponential moving average is applied to the teacher network for generating high-quality pseudo-labels. The aforementioned methods all focus on domain adaptive object detection for natural images, where the imaging mechanisms of the source and target domains are the same. Directly applying these methods to cross-modality ship detection from optical to SAR images cannot yield satisfactory results.

III. Methodology

Let $\mathcal{D}_o = \{(x_i^o, y_i^o, b_i^o)\}_{i=1}^{N_o}$ denote the optical training dataset consisted of N_o images x_i^o with the ship class indices y_i^o and their oriented bounding boxes b_i^o, and $\bar{\mathcal{D}}_s = \{(x_i^s)\}_{i=1}^{N_s}$ are the N_s training SAR images without any labeled information. Although they share the same label space, the joint probability distributions $p(\mathcal{D}_o)$ and $p(\bar{\mathcal{D}}_s, y_i^s, b_i^s)$ are not identical. Thus, when a ship detector trained on optical images is directly applied to SAR images in the inference stage, there will be a significant decline in detection performance. For mitigating such a cross-modality distribution gap, the proposed method trains a CNN with $\mathcal{D}_o$ and $\bar{\mathcal{D}}_s$ that can align the decision boundaries defined by $p(y^o, b^o|x^o)$ and $p(y^s, b^s|x^s)$. To achieve such a goal, `CroMoDa`, built upon by oriented Faster R-CNN, is mainly composed of four parts: 1) image-level feature alignment; 2) low-level feature despeckling; 3) cross-modality pseudo-label self-training; and 4) cross-modality object alignment. The overall framework of `CroMoDa` is illustrated in Fig. 2.

A. Image-Level Feature Alignment

Following the principle idea in [23], the image-level feature alignment module progressively aligns low-level and high-level features so that the backbone network can output feature maps with similar distributions when the input image modality changes. It includes two parts: low-level feature alignment and high-level feature alignment. Specifically, the low-level and high-level features of optical and SAR images extracted by the backbone network are fed into the discrimination network after passing through a gradient reversal layer. The discrimination network distinguishes whether the features belong to the optical or the SAR modality. Due to the gradient inversion, the feature extraction network can generate features with distributions that are as similar as possible. Through the adversarial interaction between the feature extraction network and the discrimination network, the backbone network is able to map the input optical images and SAR images to the same feature space, thereby obtaining intermediate-level features for the final ship detection.

1) Low-Level Feature Alignment: Image x_i and its modality label d_i are first fed into the backbone network, where they pass through the feature extraction network E_1 to obtain low-level features $\mathbf{F}_1$, i.e., $\mathbf{F}_1 = E_1(x_i)$. Then, the obtained low-level features $\mathbf{F}_1$ are input into the low-level feature discrimination network D_1, resulting in the modality category probability maps $\mathbf{P}_i^{\mathrm{L}} \in \mathbb{R}^{C\times H\times W}$. To align the low-level features obtained from optical and SAR images, we use the following loss function:

$$L_{\mathrm{img-low}} = \frac{1}{HW}\sum_{h=1}^{H}\sum_{w=1}^{W}\left[d_i\,(\mathbf{p}_i^{\mathrm{L}})^2 + (1-d_i)\,(1-\mathbf{p}_i^{\mathrm{L}})^2\right] \tag{1}$$

where H and W denote the height and width of the feature maps, respectively, and $\mathbf{p}_i^{\mathrm{L}}$ denote the channel features extracted from $\mathbf{P}_i^{\mathrm{L}}$ with the spatial indices (h, w).

2) High-Level Feature Alignment: High-level features contain rich semantic information and the modality style of images. By aligning high-level features, the backbone network can learn common semantic representations on different imagery modalities. We use the feature extraction network E_2 to obtain high-level features $\mathbf{F}_2$ from $\mathbf{F}_1$, i.e., $\mathbf{F}_2 = E_2(\mathbf{F}_1)$. Furthermore, the discrimination network outputs the modality category probabilities $\mathbf{p}_i^{\mathrm{H}} \in \mathbb{R}^{C\times 1\times 1}$ of $\mathbf{F}_2$. To achieve the feature alignment, we exploit the following loss:

$$L_{\mathrm{img-high}} = d_i\,L_{\mathrm{foc}}(\mathbf{p}_i^{\mathrm{H}}) + (1-d_i)\,L_{\mathrm{foc}}(1-\mathbf{p}_i^{\mathrm{H}}). \tag{2}$$

$L_{\mathrm{foc}}(\cdot)$ represents the focal loss function, which assigns larger weights to the images that are difficult to be classified, thereby enabling the network to better align high-level features [12].

To this end, we apply adversarial learning at the low and high levels of the backbone network, enabling us to align the optical and SAR images in terms of edges, texture features, and high-level semantic features. Accordingly, the loss for image-level

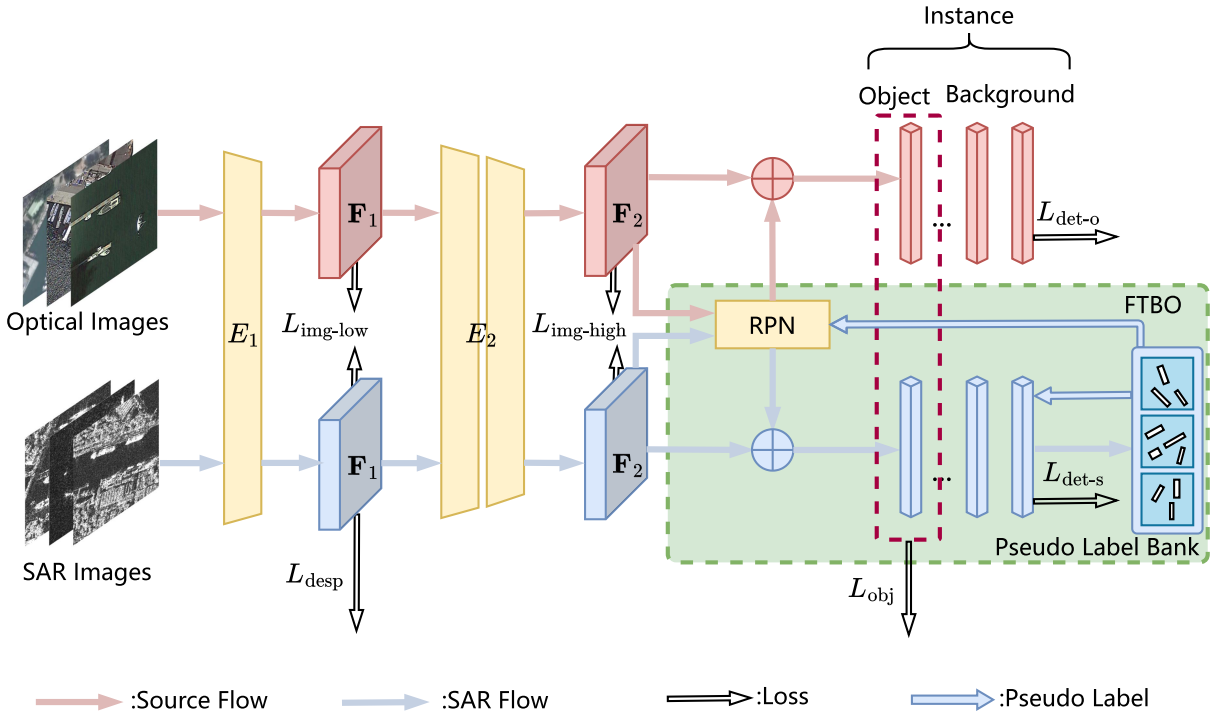
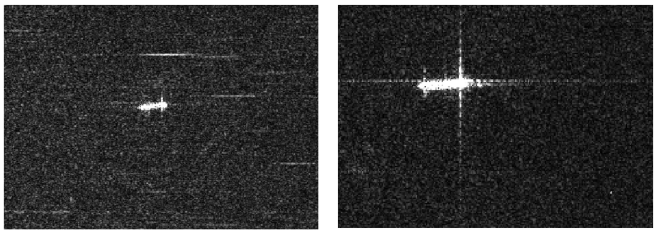


Fig. 2. Overall framework for CroMoDa.



(a) (b)

Fig. 3. (a) and (b) Speckle noise, sea clutter, and strong sidelobe effects existed in SAR images.

feature alignment is

$$L_{\text{img}} = L_{\text{img-low}} + L_{\text{img-high}}. \tag{3}$$

B. Low-Level Feature Despeckling for SAR Images

Due to the coherent processing of SAR images, sea waves often appear as clutters or speckles, which can strongly interfere with the ship interpretation. Moreover, sidelobe effects can make SAR ships appear differently with respect to the optical ones by introducing additional noise. Fig. 3(a) and (b) illustrates SAR ship examples interfered by sea clutter and sidelobe effects. Although the above image-level feature alignment can effectively align the overall features of images, it fails to achieve good alignment of ships between the two modalities at the object level. The trained optical ship detector cannot be easily adapted to identify SAR ships due to the interference unique to SAR images.

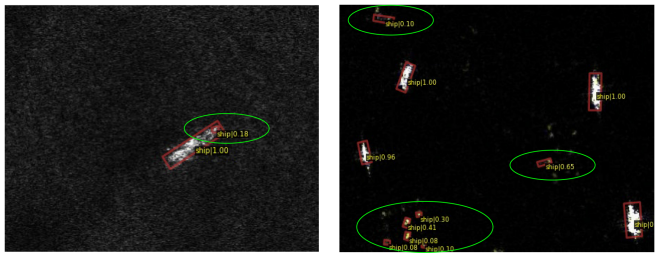
In order to avoid the potential false alarms produced by those noise, we propose a low-level feature despeckling loss term described by

$$L_{\text{desp}} = \frac{1}{HW}\sum_{h=1}^{H}\sum_{w=1}^{W}\left(\left|\mathbf{F}_1^{h+1,w} - \mathbf{F}_1^{h,w}\right| + \left|\mathbf{F}_1^{h,w+1} - \mathbf{F}_1^{h,w}\right|\right) \tag{4}$$

which constrains the total variation of the low-level features for SAR images. By minimizing L_{desp}, we would like to achieve the learned low-level features of SAR images that are not affected by the interference. It is worth noting that the despeckling loss is only carried out on low-level features rather than the high ones. Since we consider that those interference directly causes effects at the early stage of feature extraction, minimizing the total magnitude of high-level features' gradients may weaken the global semantic information.

C. Cross-Modality Pseudo-Label Self-Training for SAR Ships

The image-level feature distributions of the two modalities can be aligned as closely as possible by exploiting the above-mentioned strategies. As a result, the network can utilize the knowledge trained from optical images with labels and predict the ship locations in SAR images as well. However, due to different imaging principles, ships are characterized differently in the two imagery modalities. In SAR images, ships often appear to be composed of bright spots or strong scatterers, contrasting with the smooth and continuous objects typically observed in optical images. As shown in Fig. 4, directly exploiting the image-level aligned network to predict the ship locations in SAR images will contain lots of false alarms, due to their similar features



(a) (b)

Fig. 4. (a) and (b) Lots of false alarms present in the detection results by only considering image-level feature alignment.

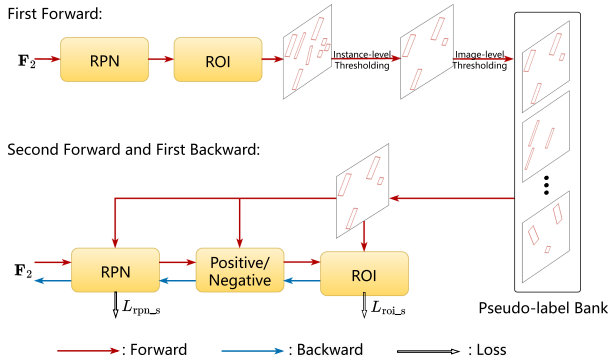


Fig. 5. Proposed FTBO strategy for cross-modality pseudo-label self-training of SAR ships.

demonstrated as the objects in optical ones. To solve such an issue, we seek to generate some supervised information in SAR images to make the network better characterize the SAR ships.

Pseudo-label self-training technology generates pseudo-labels through the prediction results of unlabeled data and uses the data containing pseudo-labels as additional samples for network training. Such a strategy further improves the generalization capability of the network, which is widely applied in semisupervised or unsupervised learning tasks [52], [53], [54]. We integrate such technology into our model to enable it to capture the unique characteristics that distinguish SAR ships from optical ones. Currently, pseudo-label self-training methods in the field of object detection are mostly based on a multistage teacher–student framework. This involves initially training the model using only source-domain data to obtain initial weights, which are then used to initialize both the teacher and student networks. During training, the teacher network is responsible for generating pseudo-labels on the target domain, and the student network utilizes these pseudo-labels for training. Such a framework has some disadvantages.

1) Due to the modality differences between optical and SAR images, the teacher network, when trained solely on optical images, generates pseudo-labels of low quality on SAR images. Consequently, it is challenging for the student network to learn modality-specific knowledge through those labels, and there is even a risk of reinforcing erroneous knowledge through training.
2) Most of the current methods are not end-to-end, and the overall performance is limited by the performance of individual learning stages.

To address the aforementioned issues, we propose an online pseudo-label self-training strategy, termed FTBO. As illustrated in Fig. 5, to obtain the losses supervised by pseudo-labels, we achieve two forward propagation processes and one backward propagation process executed on the RPN and the ROI module. During the first forward propagation, high-quality pseudo-labels are obtained through two levels of filtering, at the instance and image levels, and are stored in an offline *pseudo-label bank*. The second forward propagation retrieves pseudo-labels from the pseudo-label bank to guide the positive and negative sample allocation for candidate boxes and to calculate the loss for both RPN and the ROI module. After two forward propagations, network parameters are updated through backward propagation based on the calculated loss. Notably, the first forward propagation occurs at each training iteration, whereas the second forward propagation and backward propagation only occur when the pseudo-label bank contains labels for the input images. In detail, the proposed FTBO is introduced as follows.

1) First Forward: The SAR image features $\mathbf{F}_2$ extracted by the backbone are passed through the RPN to output candidate boxes, which then yield the predicted boxes with category probabilities and regression coordinates through the detection head and nonmaximum suppression (NMS). The category probability is used as the confidence score for the associated prediction. Generally, predicted boxes with higher confidence scores are more likely to be correctly identified. The boxes with lower ones are more likely to be background regions. Therefore, by setting a threshold θ_{ins}, most predicted boxes of background regions can be filtered out. However, when there are a large number of ships in the image with low confidence, instance-level filtering may remove these low-confidence ships. Utilizing the images with only a few pseudo-labels for training can diminish the network's detection performance. To avoid such an issue, an image-level confidence c_i is introduced to assess the overall quality of pseudo-labels within an image:

$$c_i = \frac{1}{M}\sum_{m=1}^{M} c_{i,m}\mathbb{1}_{c_{i,m}>\theta_{\text{ins}}} \tag{5}$$

where $c_{i,m}$ denotes the confidence score of the mth predicted box in x_i^s, M is the number of predicted boxes, and $\mathbb{1}$ is the indicator function.

Importantly, we exploit a pseudo-label bank $\mathbb{B} = \{x_i^s, y_i^*, b_i^*\}$ to store pseudo-labels with relatively high overall quality and the corresponding SAR images x_i^s. By setting an image-level confidence threshold θ_{img}, if the image-level confidence c_i exceeds θ_{img} and $\mathbb{B}$ does not contain pseudo-labels for that image, then the corresponding pseudo-labels for the image are stored in $\mathbb{B}$ for subsequent loss calculation. It is noteworthy that we use $\mathbb{B}$ for the following reasons.

1) The adversarial training process is not always stable. For some images, reliable pseudo-labels can be generated in the early training stage, while they may not satisfy the above criteria in the later stage. In that case, historical pseudo-labels can be retrieved from $\mathbb{B}$ to participate in training, thereby enhancing the network's training stability and convergence speed.

2) For images that cannot generate pseudo-labels, weak data augmentation can be applied during the training process. Pseudo-labels obtained from the augmented images are also stored in $\mathbb{B}$ for subsequent training of the network.

2) Second Forward and First Backward: The second forward propagation, leading to loss computation and subsequent backward propagation, commences only when the pseudo-label bank contains pseudo-labels for the current image. In that case, $\mathbf{F}_2$ is passed through the RPN to output candidate boxes. Under the guidance of pseudo-labels, these candidates are classified into positive and negative samples. By exploiting the detection head and NMS, each predicted box is associated with category probabilities and regression coordinates. For positive samples, both classification and regression losses are calculated, whereas for negative samples, only classification losses are computed. The following loss terms are exploited:

$$L_{\mathrm{RPN-s}} = L_{\mathrm{cls}}(p^{\mathrm{RPN}}_{i,m}, y^*_{i,m}) + L_{\mathrm{loc}}(b^{\mathrm{RPN}}_{i,m}, b^*_{i,m}) \tag{6}$$

$$L_{\mathrm{ROI-s}} = L_{\mathrm{cls}}(p^{\mathrm{ROI}}_{i,m}, y^*_{i,m}) + L_{\mathrm{loc}}(b^{\mathrm{ROI}}_{i,m}, b^*_{i,m}). \tag{7}$$

Here, $p_{i,m}$ and $b_{i,m}$, respectively, denote the class probability and regression parameters of the mth candidate or predicted box corresponding to the ith image, while $y^*_{i,m}$ and $b^*_{i,m}$ represent the generated pseudo-labels. L_{cls} is the cross-entropy classification loss and L_{loc} denotes the smooth L_1 loss exploited for the regression. Based on the abovementioned terms, the overall loss for SAR ship detection supervised by the generated pseudo-labels is

$$L_{\mathrm{det-s}} = L_{\mathrm{RPN-s}} + L_{\mathrm{ROI-s}}. \tag{8}$$

D. Cross-Modality Object Alignment

Considering that the basic visual appearance and structural features of objects should remain relatively consistent across different domains, existing works improve the network's generalization capability on the target domain by adopting instance-level alignment to narrow the gap between the feature representations of instances in the source and target domains [22], [26], [27], [31]. Although these methods can alleviate the instance-level feature discrepancy across domains to some extent, they face an issue where the instance-level features, extracted from the ROI based on candidate boxes in $\mathbf{F}^o_2$, encompass both the object and background features. When aligning instance features between optical and SAR images, the alignment of object instances with background instances occurs, causing the distributions of object and background features to converge. This process dilutes the unique features of ships, weakening the effectiveness of instance feature alignment. This kind of alignment is referred to as "arbitrary alignment," as shown in Fig. 6.

To address the aforementioned issue, we propose a novel instance-level alignment strategy, named cross-modality object alignment. It aims at achieving alignment only on instance features of ship regions in optical and SAR images. This strategy enables the network to more effectively extract shared features between optical and SAR ships, facilitating the transfer of knowledge learned from optical images to SAR ones at a high semantic level. Specifically, for optical images, instance features

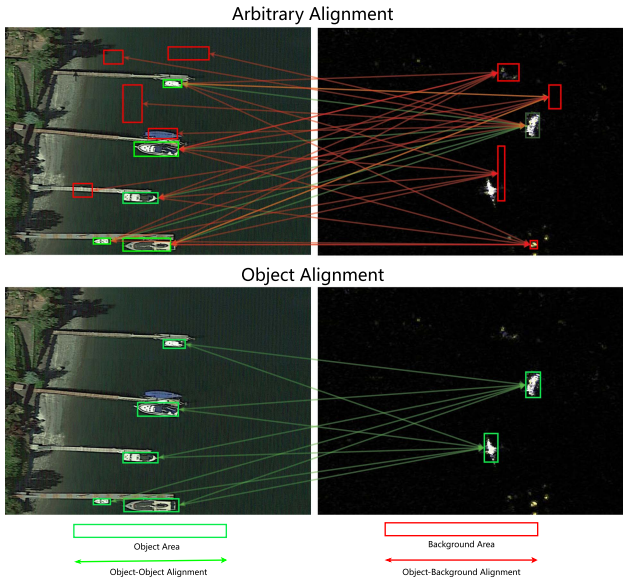


Fig. 6. Difference between arbitrary alignment of instances exploited in the other SOTAs and the proposed cross-modality object alignment.

of optical ships from the features $\mathbf{F}^o_2$ are extracted through the ROI module with accessible labels. Likewise, for SAR images, pseudo-labels from $\mathbb{B}$ are exploited to obtain instance features of SAR ships. Consistent with the instance feature fusion method proposed in [23], we obtain the low-level and high-level features of images from two discriminator networks D_1 and D_2 through the image feature alignment module. These features are then concatenated with the instance ones extracted by the ROI on $\mathbf{F}_2$ to obtain final instance features with global information. Since instance features are extracted based on ground-truth labels from optical images and high-quality pseudo-labels from SAR images, the extracted features represent the ship characteristics with a high probability, so that the alignment between the object and background regions can be avoided.

The loss calculation for the proposed cross-modality object alignment is similar to the image-level feature alignment. The obtained instance features are passed through a gradient reversal layer and then input into an instance feature discriminator network D_3, resulting in modality category probabilities $\mathbf{p}^{\mathrm{I}}_{i,j}$. i denotes the image index and j is the object index. Then, the loss term for the cross-modality object alignment is calculated by

$$L_{\mathrm{obj}} = d_i\, L_{\mathrm{foc}}(\mathbf{p}^{\mathrm{I}}_{i,j}) + (1 - d_i)\, L_{\mathrm{foc}}(1 - \mathbf{p}^{\mathrm{I}}_{i,j}). \tag{9}$$

It is worth noting that L_{obj} is obtained only when pseudo-labels for the current image exist in $\mathbb{B}$.

E. End-to-End Unsupervised Cross-Modality SAR Ship Detection

Based on the above proposed parts, the overall training for unsupervised cross-modality SAR ship detection is conducted in an end-to-end manner. The overall loss for training the network is

$$L = L_{\mathrm{det-o}} + \lambda_1 L_{\mathrm{img}} + \lambda_2 L_{\mathrm{desp}} + \lambda_3 L_{\mathrm{obj}} + \lambda_4 L_{\mathrm{det-s}} \tag{10}$$

Algorithm 1: Training Scheme for the Proposed `CroMoDa`.

Input: Optical dataset $\mathcal{D}_o$ and SAR dataset $\bar{\mathcal{D}}_s$, θ_{img}, θ_{ins}, $\mathbb{B}$, λ_1, λ_2, λ_3, λ_4
Output: `CroMoDa` network
for *epoch* **do**
  for *iteration* **do**
    Sample mini-batch $\mathbf{X}^o$, $\mathbf{X}^s$ from $\mathcal{D}_o$,$\bar{\mathcal{D}}_s$;
    Calculate the ship detection loss of optical images $L_{\text{det}-\text{o}}$;
    Calculate the image-level feature alignment loss L_{img};
    Calculate the low-level feature despeckling loss L_{desp};
    for i **do**
      Calculate image-level confidence c_i for x_i^s;
      if $c_i > \theta_{img}$ *and* b_i^*,y_i^* *not in* $\mathbb{B}$ **then**
        Save b_i^*,y_i^* to the pseudo-label bank $\mathbb{B}$;
      end
      if b_i^*,y_i^* *in* $\mathbb{B}$ **then**
        Calculate the pseudo-label self-training loss $L_{\text{det}-\text{s}}$;
        Calculate the cross-modality object alignment loss L_{obj};
      end
    end
    Update network parameters by backpropagation.
  end
end

where $L_{\text{det}-\text{o}}$ is the detection loss for optical images supervised by ground-truth labels, and $\lambda_1, \lambda_2, \lambda_3$, and λ_4 are the hyperparameters for balancing different loss terms. Algorithm 1 summarizes the training scheme for the proposed `CroMoDa`.

IV. Experiments

In this section, we first introduce the datasets used to evaluate the detection performance, the experimental setup, and the evaluation metrics adopted. We then compare our method against several current SOTA unsupervised domain adaptation detection methods on two types of optical–SAR dataset combinations. Furthermore, we conduct a series of ablation experiments on the proposed method to verify its effectiveness.

A. Experimental Setup

1) Dataset Configuration: We select DOTA [55] as the optical training dataset. SSDD [1] and RSDD [56] are the two SAR datasets used for unsupervised ship detection. The DOTA dataset is currently one of the mainstream optical remote sensing datasets for object detection, with images sourced from Google Earth and two Chinese satellites (Gaofen-2 and Jilin-1). This dataset contains a total of 2806 images, with sizes ranging from 800 × 800 to 4000 × 4000 pixels, and includes 188 282 instances across 15 categories, such as airplanes and ships, using oriented annotations. We first obtain 31 879 images of size 1024 × 1024 through slicing and then select 1869 images containing ships as the training dataset. The SSDD dataset is composed of 1160 SAR images acquired from three SAR satellites: Radarsat-2, TerraSAR-X, and Sentinel-1. Those images have an average size of 500 × 500, containing a total of 2456 oriented ships. The image resolution ranges from 1 to 15 m per pixel, featuring ships of various sizes and materials in both nearshore and offshore scenes. We divide the SSDD dataset into 928 training images and 232 test images with an 8:2 split ratio without overlap and repeat the division process three times to obtain three train-test datasets for reliable evaluation. The RSDD-SAR dataset is composed of 127 scenes and 7000 SAR images of size 512 × 512, which contains a total of 10263 oriented ships. We select 1600 and 400 images for training and testing with the same split ratio, and repeat three times as above.

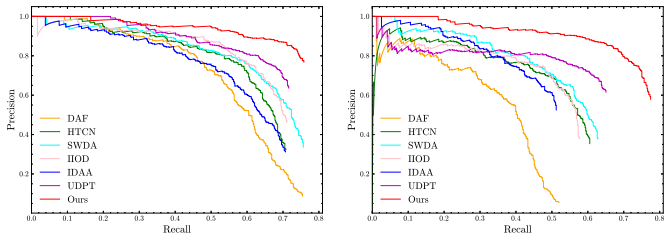


Fig. 7. PR curves of all the considered methods on (a) DOTA → SSDD and (b) DOTA → RSDD.

2) Implementation Details: During the training phase, images from both modalities are resized to a dimension where the longer side is 1024 pixels, maintaining the aspect ratio of the images. In addition, random horizontal flipping is applied. In the testing phase, SAR images are resized in the same manner without any data augmentation. We utilize the stochastic gradient descent optimizer to train the network, with an initial learning rate of 0.005, a weight decay of 0.0001, and a momentum of 0.9. The learning rate is reduced by a factor of 0.1 at the 32nd and 44th epochs. The network is trained for a total of 48 epochs, with a batch size of four training images, including two optical and two SAR images. All the implementations are based on MMRotate [57] and carried out on a server with an NVIDIA RTX3090 GPU. The hyperparameter settings are $\theta_{\text{ins}} = 0.5$, $\theta_{\text{img}} = 0.97$, $\lambda_1 = 0.5$, $\lambda_2 = 1.0$, $\lambda_3 = 0.2$, and $\lambda_4 = 1.0$. During the testing phase, detected boxes with an IoU greater than 0.5 compared with the ground-truth boxes are considered to be correct detection. To quantitatively assess our method, we use the following four metrics:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{11}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{12}$$

$$\text{F}_1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{13}$$

$$\text{mAP} = \frac{1}{m} \sum_{i=1}^{m} \int P(r) dr \tag{14}$$

where TP, FP, and FN represent true positive, false positive, and false negative, respectively. Precision is the proportion of

TABLE I
QUANTITATIVE ANALYSIS OF ALL THE CONSIDERED METHODS ON DOTA → SSDD AND DOTA → RSDD

Method	DOTA → SSDD				DOTA → RSDD			
	Precision	Recall	F_1	mAP	Precision	Recall	F_1	mAP
R-Faster RCNN [16] (optical)	$33.5_{\pm 21.9}$	$48.6_{\pm 9.7}$	$33.6_{\pm 17.1}$	$39.4_{\pm 6.6}$	$41.6_{\pm 20.7}$	$39.8_{\pm 5.0}$	$37.1_{\pm 14.6}$	$32.9_{\pm 4.1}$
R-Faster RCNN [16] (supervised)	$88.8_{\pm 1.4}$	$95.1_{\pm 0.5}$	$91.8_{\pm 1.0}$	$90.6_{\pm 0.1}$	$78.1_{\pm 4.0}$	$90.7_{\pm 1.1}$	$83.9_{\pm 2.5}$	$86.1_{\pm 4.1}$
DAF [22] (2018)	$14.1_{\pm 7.6}$	$73.9_{\pm 2.8}$	$22.8_{\pm 9.5}$	$58.1_{\pm 3.2}$	$14.5_{\pm 12.0}$	$51.6_{\pm 3.3}$	$20.5_{\pm 13.4}$	$38.4_{\pm 3.4}$
SWDA [23] (2019)	$42.6_{\pm 7.4}$	$75.0_{\pm 2.5}$	$53.9_{\pm 5.4}$	$64.0_{\pm 1.3}$	$33.6_{\pm 7.8}$	$63.5_{\pm 3.3}$	$43.3_{\pm 6.1}$	$52.9_{\pm 2.9}$
HTCN [26] (2020)	$36.8_{\pm 15.9}$	$72.2_{\pm 2.2}$	$46.5_{\pm 17.5}$	$61.9_{\pm 3.1}$	$35.7_{\pm 7.9}$	$62.5_{\pm 2.8}$	$44.9_{\pm 6.9}$	$51.5_{\pm 2.4}$
IIOD [27] (2022)	$38.1_{\pm 11.7}$	$71.8_{\pm 5.0}$	$48.4_{\pm 10.6}$	$61.1_{\pm 3.1}$	$24.8_{\pm 14.3}$	$58.9_{\pm 5.3}$	$32.5_{\pm 14.9}$	$47.5_{\pm 5.2}$
UDPT [30] (2022)	$\mathbf{70.6}_{\pm 6.5}$	$69.8_{\pm 4.1}$	$70.0_{\pm 3.5}$	$63.7_{\pm 4.0}$	$52.9_{\pm 13.8}$	$59.3_{\pm 8.6}$	$55.4_{\pm 11.2}$	$48.6_{\pm 7.8}$
IDAA [31] (2023)	$19.5_{\pm 6.4}$	$71.3_{\pm 2.1}$	$30.1_{\pm 7.9}$	$57.1_{\pm 1.6}$	$24.2_{\pm 15.0}$	$47.0_{\pm 2.2}$	$29.6_{\pm 14.8}$	$37.7_{\pm 3.5}$
CroMoDa	$68.4_{\pm 9.1}$	$\mathbf{78.7}_{\pm 4.3}$	$\mathbf{72.7}_{\pm 4.0}$	$\mathbf{72.7}_{\pm 3.7}$	$\mathbf{65.6}_{\pm 6.2}$	$\mathbf{73.7}_{\pm 3.0}$	$\mathbf{69.3}_{\pm 3.8}$	$\mathbf{66.2}_{\pm 1.5}$

The mean and standard deviation values are calculated on a basis of nine runs.
Bold highlights the highest performances in all the considered methods.

TABLE II
COMPARISON OF INFERENCE SPEED, PARAMETER COUNT, AND FLOATING-POINT OPERATION COUNT

Method	FPS	Params (M)	Flops (G)
R-Faster RCNN [16]	**34.0**	**85.46**	**177.51**
DAF [22] (2018)	32.4	87.18	**177.51**
SWDA [23] (2019)	26.6	92.33	190.26
HTCN [26] (2020)	27.6	94.23	199.26
IIOD [27] (2022)	11.7	148.6	560.75
UDPT [30] (2022)	**34.0**	**85.46**	**177.51**
IDAA [31] (2023)	33.2	87.18	**177.51**
CroMoDa	26.4	92.44	190.26

Image size is 1024 × 1024 during testing for model inference speed and floating-point operation count.
Bold highlights the highest performances in all the considered methods.

actual positives in samples predicted as positive by the network, while Recall is the proportion of actual positives that the network correctly predicts as positive among all true positives. The F1 score is the harmonic mean of Precision and Recall, offering a composite measure of the model's performance. $P(\cdot)$ represents Precision as a function of Recall, and integrating this over the range gives the average precision (AP) for a category. Averaging the AP values across multiple categories yields the mAP, which is used to comprehensively evaluate the network's performance. It is worth noting that each of our experiments is conducted on three different sets of train–test splits, with each run being randomized three times for each set. The final experimental results are obtained by taking the average of these nine runs.

3) Baselines: To evaluate the effectiveness of the proposed method, we compare it with several SOTA object detection methods: 1) *R-Faster RCNN [16]*; in the experiments, we exploit *R-Faster RCNN (optical)* to represent the approach of using only labeled optical images for training the network and testing it on SAR images, while *R-Faster RCNN (supervised)* represents training and testing the network on SAR images in a fully supervised manner; 2) *DAF [22]*; 3) *SWDA [23]*; 4) *HTCN [26]*; 5) *IIOD [27]*; 6) *UDPT [30]*; and 7) *IDAA [31]*.

B. Experimental Results

1) Comparison to SOTA Methods: To validate the effectiveness of the proposed method, we conduct quantitative and qualitative experiments compared with the abovementioned SOTAs on two dataset configurations: DOTA → SSDD and DOTA → RSDD. Table I demonstrates the quantitative results of all the considered methods, where the mean and standard deviation values are calculated on a basis of nine runs. It can be observed that the proposed method achieves the best detection performance of all. For DOTA → SSDD, CroMoDa's F_1 and mAP scores, respectively, improve around 3% and 9% compared with the second-best method. For DOTA → RSDD, the proposed method significantly increases the detection accuracy by around 13% F_1 score and 14% mAP score compared with the second best. In addition, our method maintains high precision while achieving the highest recall. Since pseudo-labels of SAR ships are considered for supervising the network learning in both UDPT and CroMoDa, they demonstrate qualitative improvement in detection performances on both datasets compared to the others. Such improvement is attributed to the suppression of false alarms by utilizing the pseudo-label self-training strategy. Table I shows a considerable standard deviation in some results, which can be attributed to two primary factors. First, both the proposed method and the comparative methods employ adversarial training within the network. The adversarial nature between the feature extraction network and the discriminator network can lead to instability during training, resulting in significant variations in the outcomes of each model training session. Second, we perform three random splits of the training and testing sets, and different splits led to substantial differences in the accuracy of the trained models. Table II presents a comparative analysis of our proposed method with other methods in terms of FPS, parameter count, and floating-point operations. It is evident that our method aligns closely with most methods in terms of inference speed, parameter count, and floating-point operations, with notable differences observed compared to DAF, UDPT, and IDAA. However, our method significantly outperforms others in terms of detection accuracy. It is noteworthy that UDPT exhibits identical metrics to R-Faster RCNN because UDPT is a three-stage algorithm. The third-stage model is identical to R-Faster RCNN and is exclusively used for inference. IDAA introduces an imbalanced prediction consistency loss on top of DAF, resulting in identical parameter counts. The number of floating-point operations for IDAA, UDPT, DAF, and R-Faster RCNN is the same because we delete part of the network and loss calculation during inference. Fig. 7 displays the Precision–Recall (PR) curves of all the considered methods, with Fig. 7(a) and (b) corresponding to the results on the SSDD and RSDD

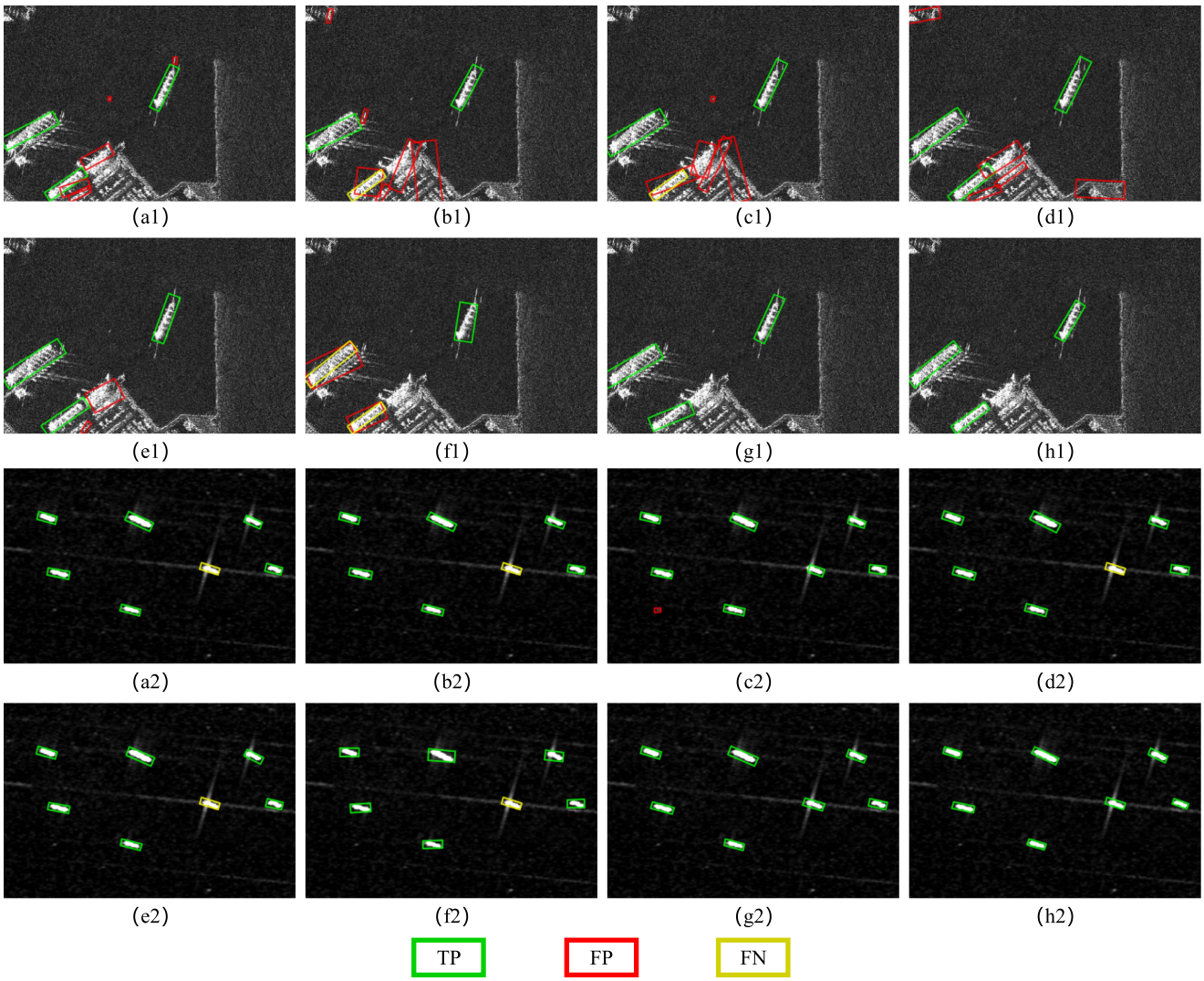


Fig. 8. Visualized detection results of 1) nearshore scene and 2) offshore scene selected from SSDD test images. (a1) and (a2) DAF. (b1) and (b2) HTCN. (c1) and (c2) SWDA. (d1) and (d2) IIOD. (e1) and (e2) IDAA. (f1) and (f2) UDPT. (g1) and (g2) `CroMoDa`. (h1) and (h2) Ground truth. TP, FP, and FN are annotated in different colors.

datasets, respectively. The area under the PR curve, enclosed by the curve and the x-axis, represents the AP value, providing an intuitive indication of the performance. The closer the curve is to the upper right corner of the coordinate axis, the better the detection performance. It can be observed that our curve is almost positioned above all the others, fully demonstrating the superiority of our method. Our method's curve is almost a horizontal line in the upper left corner, while the others show a process of steep decline followed by a rise. This indicates that the predicted boxes of our method with high confidence are very accurate, whereas other methods tend to predict backgrounds as ships and assign high confidence to these incorrect predictions.

In terms of the qualitative experiments, Fig. 8 shows the visualized detection results of our method compared with the other SOTA methods on the SSDD dataset, including both nearshore and offshore scenes. In the nearshore scene, although DAF, IIOD, and IDAA can correctly detect all the ships, they inevitably produce a large number of false alarms due to the presence of many areas on land similar to ships. In addition, while HTCN, SWDA, and UDPT provide predicted boxes for all ship areas, some of them have a small IoU with reference to the ground truths and are deemed false alarms. Compared with the others, our method successfully predicts all the ships and suppresses the occurrence of false alarms, achieving the best detection result . In the offshore scene, strong sidelobe interference creates a significant difference in the appearance of some ships, leading to false negatives via DAF, HTCN, IIOD, IDAA, and UDPT. Although SWDA successfully detects ships with sidelobe interference, its performance is affected by sea clutter, resulting in false alarms. Our method benefits from the use of despeckling on low-level features, which filters out a significant amount of sea clutter, weakens the impact of sidelobe interference on detection, and outperforms the others. Fig. 9 shows the results for the RSDD dataset. In Scene 1, DAF, HTCN, SWDA, and IIOD all experience a large number of false alarms, and DAF's prediction boxes almost cover the entire land area on the left and right of the image. In addition, DAF, HTCN, IDAA, and UDPT all fail to predict the ship located at the bottom of the

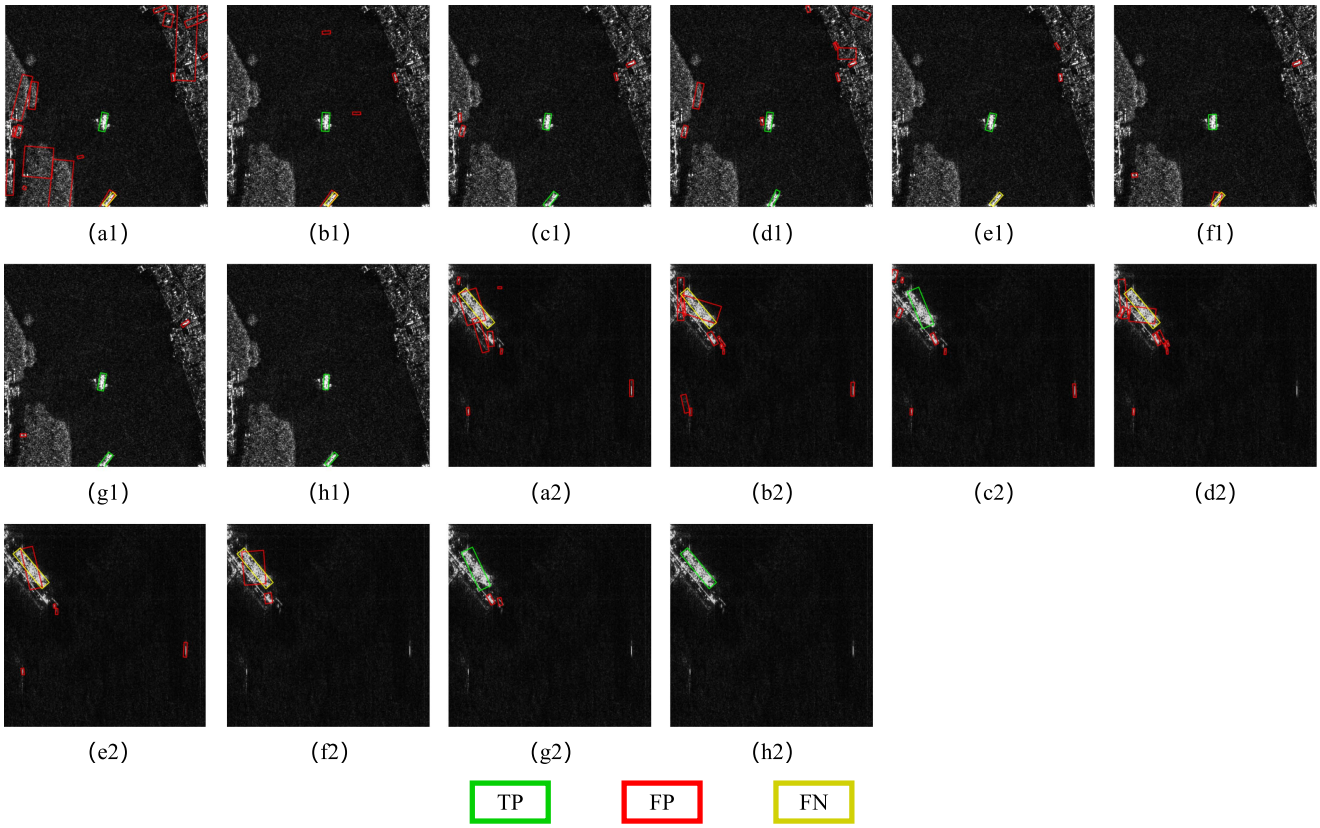


Fig. 9. Visualized detection results of two nearshore scenes selected from RSDD test images. (a1) and (a2) DAF. (b1) and (b2) HTCN. (c1) and (c2) SWDA. (d1) and (d2) IIOD. (e1) and (e2) IDAA. (f1) and (f2) UDPT. (g1) and (g2) CroMoDa. (h1) and (h2) Ground truth. TP, FP, and FN are annotated in different colors.

TABLE III
ABLATION STUDY OF THE PROPOSED METHOD ON THE SSDD TEST SET

Method	L_{desp}	$\mathbb{B}$	L_{obj}	Precision	Recall	F_1	mAP
Baseline				$42.6_{\pm 7.4}$	$75.0_{\pm 2.5}$	$53.9_{\pm 5.4}$	$64.0_{\pm 1.3}$
Ours	✓			$45.2_{\pm 9.6}$	$77.9_{\pm 4.4}$	$56.4_{\pm 6.6}$	$68.7_{\pm 2.8}$
		✓		$66.5_{\pm 11.4}$	$77.2_{\pm 3.1}$	$70.8_{\pm 6.8}$	$70.6_{\pm 3.7}$
		✓	✓	$\mathbf{71.3}_{\pm 6.5}$	$78.2_{\pm 3.5}$	$\mathbf{74.4}_{\pm 3.6}$	$72.3_{\pm 3.8}$
	✓	✓	✓	$68.4_{\pm 9.1}$	$\mathbf{78.7}_{\pm 4.3}$	$72.7_{\pm 4.0}$	$\mathbf{72.7}_{\pm 3.7}$

Bold highlights the highest performances in all the considered methods.

scene. Although DAF, HTCN, and UDPT can correctly identify the location of the ships, inaccuracies in the size and orientation of their predicted boxes led to false negatives. In Scene 2, DAF, HTCN, SWDA, IIOD, and IDAA, to some degrees, predict sea clutter as ships. Due to the influence of land, the predicted boxes from DAF, HTCN, IIOD, IDAA, and UDPT deviated from the ground truths, leading to false negatives. In contrast, our method achieves the best predicted performance in both scenes. Fig. 10 illustrates the pseudo-labels generated during the training process by our proposed method and UDPT. Fig. 10(a), (b) and (c), (d) represent nearshore and offshore scenes, respectively. It can be observed that the proposed method generates high-quality pseudo-labels in both scenes. This is attributed to the fact that we have incorporated image-level filtering on top of instance-level filtering, which helps in avoiding the model's misguidance by noisy labels during the early stages of training. In addition, the introduced pseudo-label bank helps in mitigating the cumulative errors caused by noisy pseudo-labels during the training process.

2) Ablation Study: Tables III and IV correspond to the results of the ablation studies on the two dataset configurations, respectively. In the tables, "Baseline" refers to R-Faster RCNN with image-level feature alignment. It can be observed that each component of our method contributes to enhancing the network's performance, and their combined effects enable the network to achieve the best result. By adding L_{desp}, we observe an increase in both precision and recall values on the two datasets. This improvement is attributed to L_{desp} reducing background noise and suppressing sidelobe interference, which decreases false alarms and enhances the detection rate of SAR ships. The pseudo-label self-training technique significantly boosts accuracy because it treats a large number of low-confidence ships as background,

TABLE IV
ABLATION STUDY OF THE PROPOSED METHOD ON THE RSDD TEST SET

Method	L_{desp}	$\mathbb{B}$	L_{obj}	Precision	Recall	F_1	mAP
Baseline				$33.6_{\pm 7.8}$	$63.5_{\pm 3.3}$	$43.3_{\pm 6.1}$	$52.9_{\pm 2.9}$
Ours	✓			$36.6_{\pm 7.3}$	$66.0_{\pm 2.9}$	$46.8_{\pm 5.7}$	$55.1_{\pm 2.7}$
		✓		$62.1_{\pm 11.4}$	$68.0_{\pm 3.0}$	$64.2_{\pm 6.1}$	$59.3_{\pm 2.3}$
		✓	✓	$60.6_{\pm 5.2}$	$70.1_{\pm 3.1}$	$64.8_{\pm 3.1}$	$63.2_{\pm 3.6}$
	✓	✓	✓	$\mathbf{65.6}_{\pm 6.2}$	$\mathbf{73.7}_{\pm 3.0}$	$\mathbf{69.3}_{\pm 3.8}$	$\mathbf{66.2}_{\pm 1.5}$

Bold highlights the highest performances in all the considered methods.

TABLE V
EFFECTIVENESS STUDY OF THE PROPOSED L_{DESP} IMPOSED ON DIFFERENT STAGES

	DOTA → SSDD				DOTA→ RSDD			
	Precision	Recall	F_1	mAP	Precision	Recall	F_1	mAP
w/o L_{desp}	$42.6_{\pm 7.4}$	$75.0_{\pm 2.5}$	$53.9_{\pm 5.4}$	$64.0_{\pm 1.3}$	$33.6_{\pm 7.8}$	$63.5_{\pm 3.3}$	$43.3_{\pm 6.1}$	$52.9_{\pm 2.9}$
image despeckling	$\mathbf{47.9}_{\pm 7.0}$	$75.4_{\pm 2.6}$	$\mathbf{58.4}_{\pm 5.3}$	$65.9_{\pm 1.9}$	$38.4_{\pm 8.7}$	$62.5_{\pm 1.7}$	$47.0_{\pm 6.7}$	$53.2_{\pm 2.2}$
low-level feature despeckling	$45.2_{\pm 9.6}$	$\mathbf{77.9}_{\pm 4.4}$	$56.4_{\pm 6.6}$	$\mathbf{68.7}_{\pm 2.8}$	$\mathbf{45.0}_{\pm 10.3}$	$\mathbf{65.4}_{\pm 3.3}$	$\mathbf{52.5}_{\pm 6.3}$	$\mathbf{56.2}_{\pm 1.4}$
high-level feature despeckling	$44.6_{\pm 13.6}$	$67.3_{\pm 4.7}$	$52.2_{\pm 8.9}$	$59.3_{\pm 2.4}$	$42.9_{\pm 6.6}$	$65.2_{\pm 1.9}$	$51.5_{\pm 4.7}$	$55.3_{\pm 1.7}$

Bold highlights the highest performances in all the considered methods.

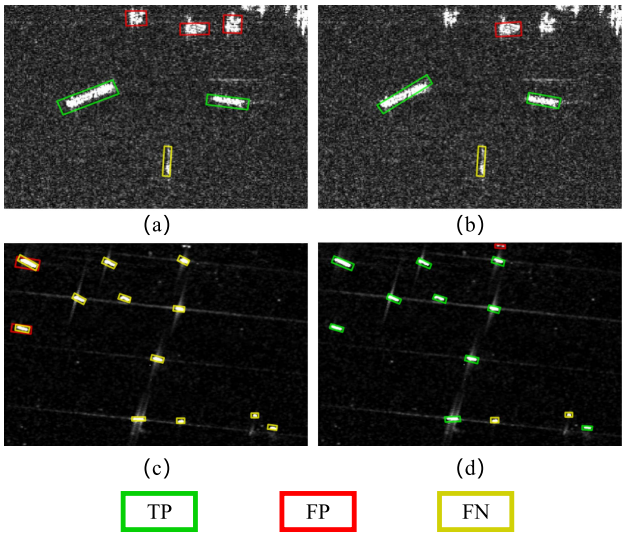


Fig. 10. Visualization results of pseudo-labels. (a) and (c) UDPT. (b) and (d) Ours.

with their confidence levels gradually approaching zero during training. Cross-modality object alignment, by aligning the features of the ship regions, makes their features in optical and SAR images more similar, effectively improving the network's accuracy and recall rates.

a) Effect of L_{desp}: In Table V, we investigate different detection performances when L_{desp} is imposed at different stages, i.e., image despeckling, low-level feature despeckling, and high-level feature despeckling. Moreover, the detection result without using L_{desp} is also reported. It can be observed that L_{desp} conducted on low-level features can achieve the best detection accuracy compared to the others. Since the speckle effect belongs to image-level characteristics within SAR images, high-level features extracted by the backbone are mainly semantic and are less affected by the speckle. Thus, L_{desp} imposed on high-level features does not exhibit any performance improvement. Furthermore, Fig. 11 demonstrates two detection results and the associated low-level features without and with L_{desp}, where Fig. 11(a) and (b) corresponds to the effects of sea clutter and sidelobe interference. On the left side of the subfigures, it can be observed that both sea clutter and sidelobe interference can lead to much noise over the background areas of the low-level features. This results in the occurrence of false alarms in the detection results, and the predicted boxes for ships also exhibit deviations. As comparison, the minimization of L_{desp} conducted on low-level features can lead to much smoother background features, which better dominate the appearance of ships. In return, the locations of ships can be more precisely predicted.

b) Effect of object alignment: In order to analyze the effect of the proposed object alignment method, we compare the instance-level similarities for the inter- and intra-modality cases with respect to the method proposed in SWDA. Fig. 12(a) illustrates the instance similarities across the DOTA dataset and the SSDD dataset, while Fig. 12(b) shows the self-similarity of instance features within the SSDD dataset. Likewise, Fig. 12(c) and (d) presents the same results for the RSDD dataset. Observed from the results, our method can increase the similarity between instance features in optical images and SAR images by comparing to SWDA. This improvement is attributed to our proposed cross-modality object alignment strategy, which aligns the ship features, thereby narrowing the gap in feature distribution between ship areas in optical and SAR images. Moreover, according to Fig. 12(b) and (d), our method increases instance similarity among ships in SAR images, which is due to the use of a pseudo-label self-training strategy during the training process. It enables the network to learn more distinctive features specific to SAR ships.

c) Effect of $\mathbb{B}$: Fig. 13 demonstrates the impact of the proposed pseudo-label bank on the network's performance in the test set of DOTA → SSDD. In addition, we compare the performance with the other two cases: 1) without using a bank, denoted as w/o Bank; and 2) Bank updating. Compared to the proposed strategy, Bank updating refers to the strategy in which $\mathbb{B}$ is updated in each iteration. By comparing ours with w/o Bank, using a pseudo-label bank in self-training can accelerate the network convergence speed and significantly improve the

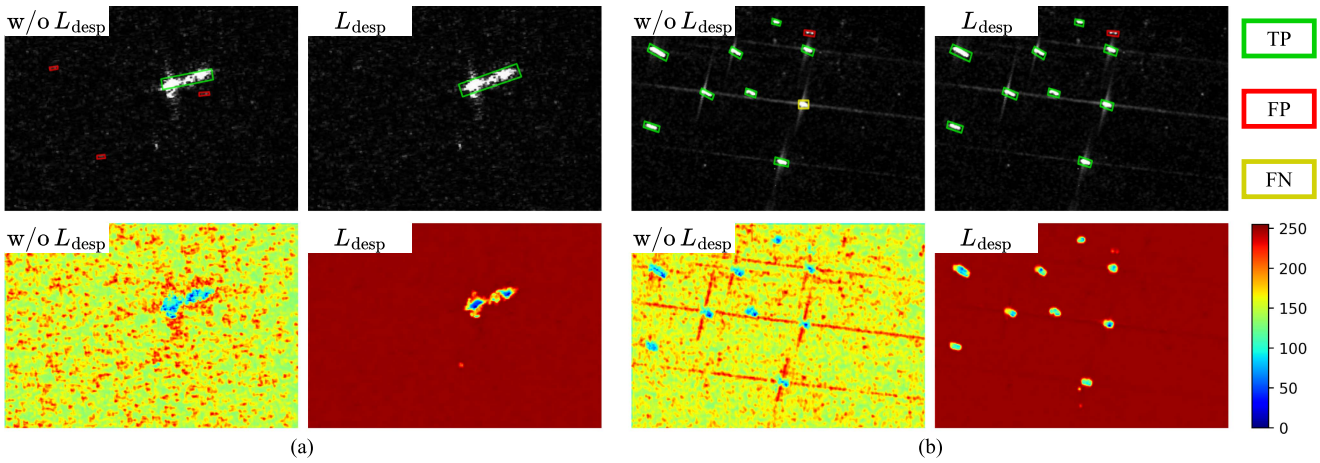


Fig. 11. Impact of L_{desp} on the detection results and low-level features, where (a) and (b) correspond to the effects of sea clutter and sidelobe interference.

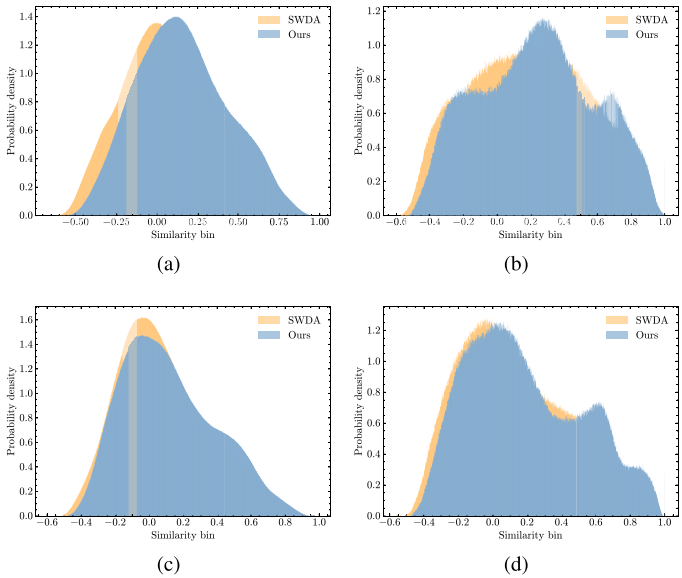


Fig. 12. Ship feature similarity distributions of (a) DOTA ↔ SSDD, (b) SSDD ↔ SSDD, (c) DOTA ↔ RSDD, and (d) RSDD ↔ RSDD.

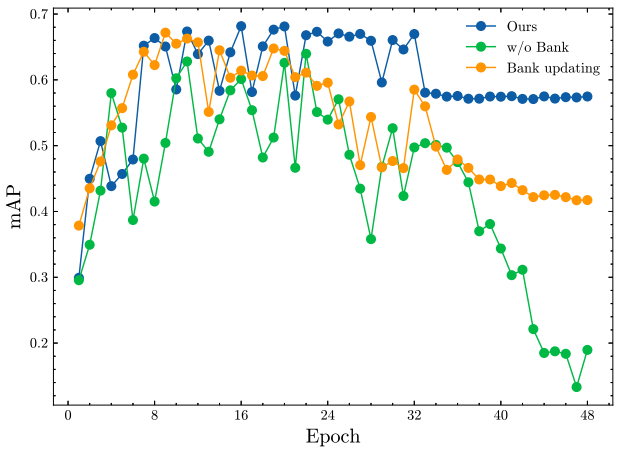


Fig. 13. Impact of applying different pseudo-label bank strategies on the network's detection performances during self-training.

network's precision. This improvement is because w/o Bank directly uses pseudo-labels generated in the current batch for network training. However, the adversarial training of the network is usually unstable during the training process. Images that can generate pseudo-labels in one training iteration may not be able to produce suitable pseudo-labels in the next, resulting in fewer pseudo-labels being involved in training. By contrast, our pseudo-label bank can store suitable pseudo-labels, ensuring that the training stage can proceed with these labels even when new suitable pseudo-labels cannot be generated. It can be seen that the performance of w/o Bank continues to decline during the later stages of training. A possible reason is that w/o Bank directly uses pseudo-labels generated in the current batch to update the network, and the next batch uses the updated network to generate pseudo-labels for training. In cases where pseudo-labels contain noise, the error may gradually amplify as training progresses, leading to a continuous decline in network performance [58]. To verify the above speculation, we compare Bank updating with our method. It can be observed that in the early stages of training, the performance of Bank updating is almost identical to our method. However, in the middle and later stages, as the pseudo-labels are updated, the errors in the pseudo-labels gradually lead to a decline in performance. The learning curve falls between ours and w/o Bank, indicating that while Bank updating helps mitigate some of the instability, it still suffers from the accumulation of errors over time, unlike our method, which maintains higher stability and accuracy throughout the training process.

3) Discussion: As mentioned earlier, we conduct extensive experiments from different perspectives to verify the effectiveness of CroMoDa. Many existing cross-modality object detection methods have shown excellent performance on data within the same modality, but they generally perform poorly on the highly challenging task of cross-modality transfer from optical to SAR. Compared to optical images, SAR images contain speckle noise and sidelobe interference, which can affect the intensity consistency of ship objects, resulting in a significant number of missed detections. In addition, the speckle noise

in land areas resembles ship objects in appearance, leading to numerous false alarms. To the best of our knowledge, current cross-modality object detection algorithms have not considered the impact of noise on cross-modality results. Therefore, we add a denoising loss to the backbone network to mitigate the influence of noise on features and conduct experiments to evaluate the effect of adding the denoising loss at different stages of the backbone network. Based on the obtained results, we believe that using denoising loss on low-level features is optimal. A reasonable explanation is that speckle noise is a local feature of the underlying image, and denoising at lower levels can prevent interference with higher level features. Moreover, high-level features represent global semantic features, and using denoising loss on them can weaken the global semantic features.

Most current cross-modality object detection methods incorporate adversarial training in the model to learn modality-invariant feature representations, known as transferability. However, achieving transferability comes at a cost: adversarial training may compromise the discriminability of the features. Discriminability refers to the model's ability to locate and distinguish different instances [26]. In the cross-modality ship object detection task from optical to SAR, this manifests as a reduced ability of the model to differentiate between ships and the background, as well as decreased localization accuracy. We introduce pseudo-label self-training in the model, generating pseudo-labels for SAR images to perform supervised training, thereby enhancing the discriminability of features on SAR images. Furthermore, we find in experiments that the quality of pseudo-labels directly affects the model's performance. Simple instance-level filtering can introduce a large number of noisy labels, but adding image-level filtering can remove noisy pseudo-labels. In addition, generating pseudo-labels based on classification scores does not guarantee localization accuracy, leading to localization errors. Current pseudo-label self-training methods update pseudo-labels during training, causing cumulative errors as the model iterates. We introduce a pseudo-label bank to avoid cumulative errors caused by multiple updates of pseudo-labels.

V. Conclusion

In this article, we propose a cross-modality SAR ship detection method by applying the knowledge learned from labeled optical images in an unsupervised manner. Experimental results on two optical–SAR dataset configurations demonstrate that our method significantly outperforms other SOTA methods in detection accuracy with an mAP score greater than 10%, proving its effectiveness and superiority in cross-modality SAR ship detection. In the future, we plan to explore lightweight cross-modality SAR ship detection methods to achieve real-time applications.

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